

**Supporting Information for Valence to Core X-ray Emission
Spectroscopy: A Sensitive Probe of the Nature of a Bound
Ligand**

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Sample ORCA Geometry Optimization + XES Calculation Input File

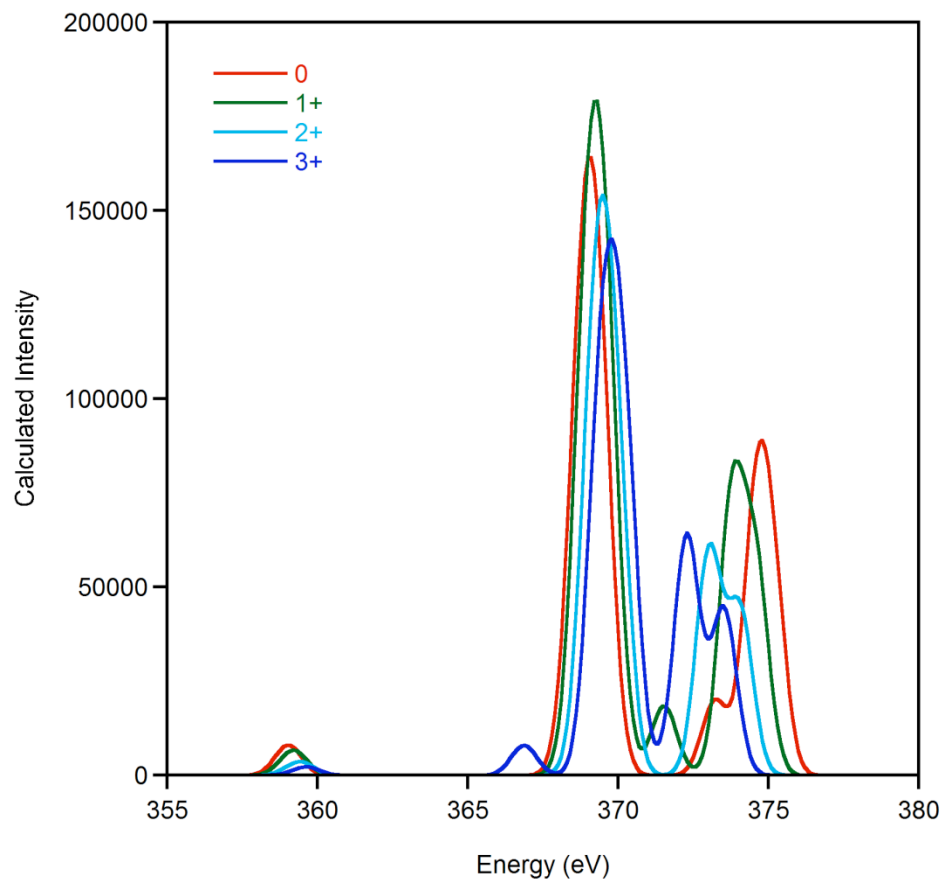
```
! UKS BP86 TZVP def2-TZV/J TightSCF SlowConv SCFConv7 COSMO
! Grid4 NoFinalGrid Normalprint
! PAL8
! OPT
```

```
%scf
    MaxIter 500
end
```

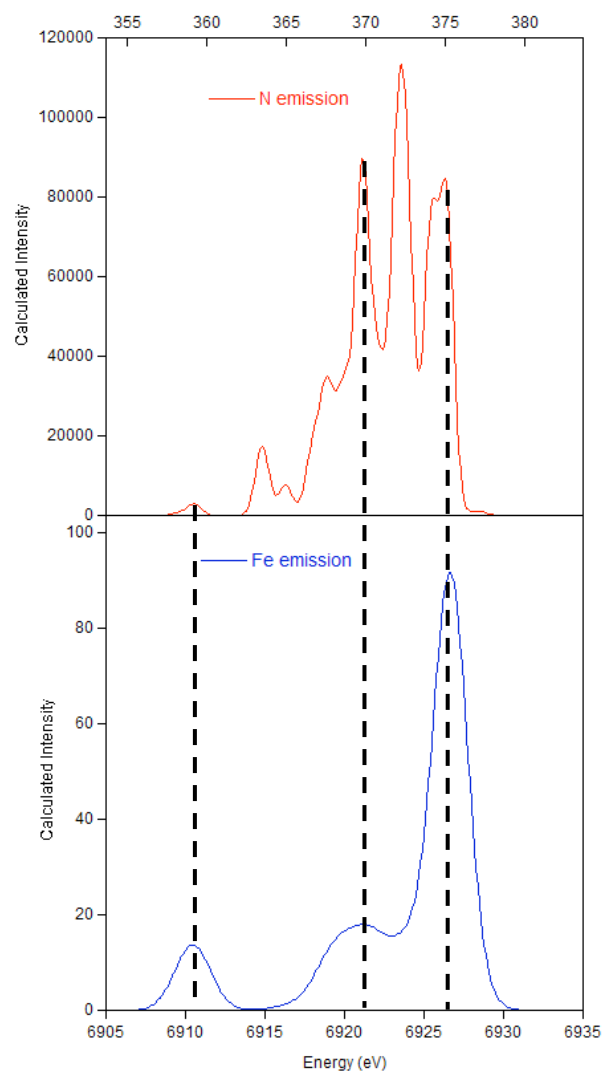
```
%maxcore 1024
```

```
%xes CoreOrb 0,0
    OrbOp 0,1
end
```

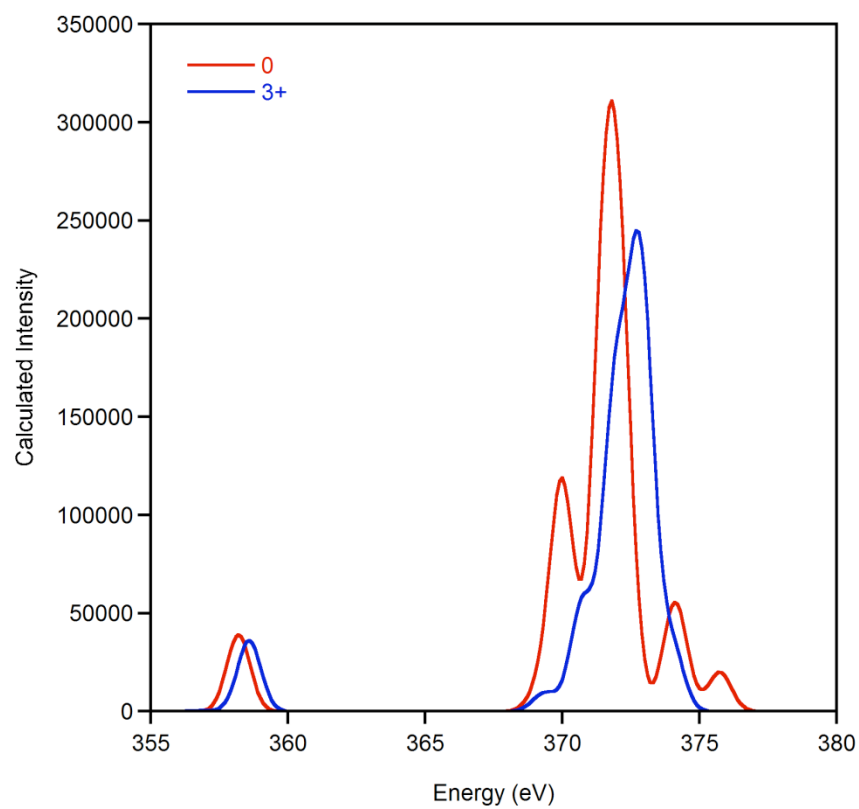
```
* xyz 3 6
Fe      0.00000000    0.00000000    0.00000000
N      -2.00000000    0.00000000    0.00000000
H      -2.42050000    0.00220000   -0.93680000
H      -2.42190000   -0.81130000    0.46830000
H      -2.42100000    0.81130000    0.46950000
N       2.00000000    0.00000000    0.00000000
H       2.42050000   -0.00220000   -0.93680000
H       2.42190000    0.81130000    0.46830000
H       2.42100000   -0.81130000    0.46950000
N       0.00000000    2.00000000    0.00000000
H      -0.00220000    2.42050000   -0.93680000
H       0.81130000    2.42190000    0.46830000
H      -0.81130000    2.42100000    0.46950000
N       0.00000000   -2.00000000    0.00000000
H      -0.00220000   -2.42050000   -0.93680000
H       0.81130000   -2.42190000    0.46830000
H      -0.81130000   -2.42100000    0.46950000
N       0.00000000    0.00000000   -2.00000000
H      -0.00220000   -0.93680000   -2.42050000
H       0.81130000    0.46830000   -2.42190000
H      -0.81130000    0.46950000   -2.42100000
N       0.00000000    0.00000000    2.00000000
H      -0.00220000   -0.93680000    2.42050000
H       0.81130000    0.46830000    2.42190000
H      -0.81130000    0.46950000    2.42100000
*
```



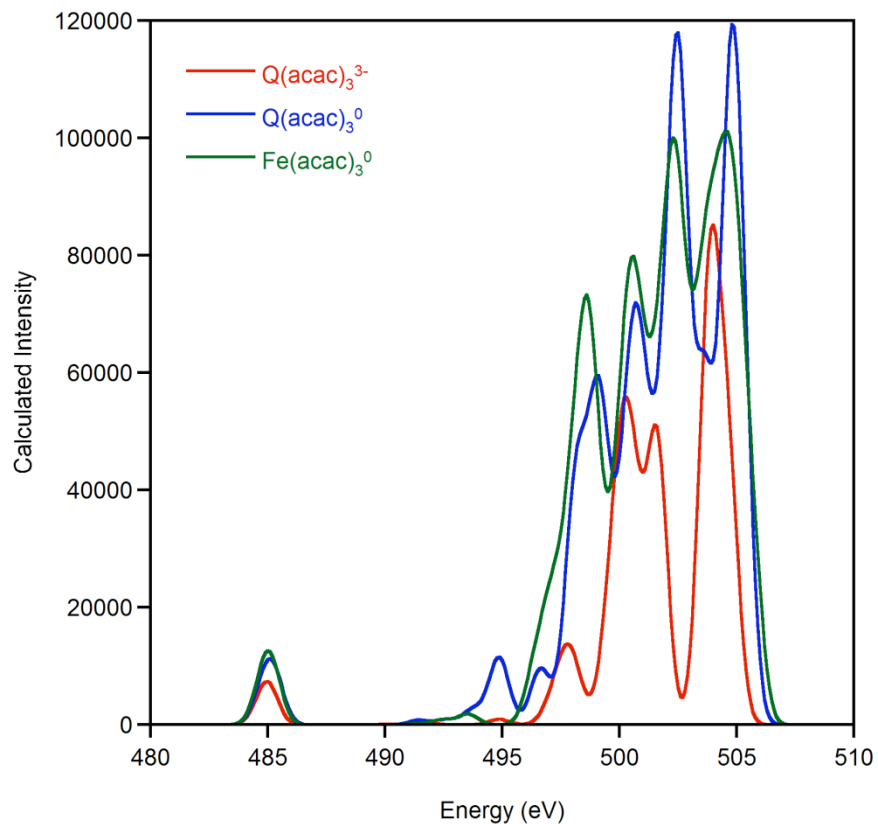
S1. The effect on the N emission spectra from increasing charge of Q from 0 to +3 for the hypothetical construct $Q(NH_3)_6^{n+}$ is shown. Note the slight destabilization of the N 2s and N-H bond features and the strong stabilization of the lone pair.



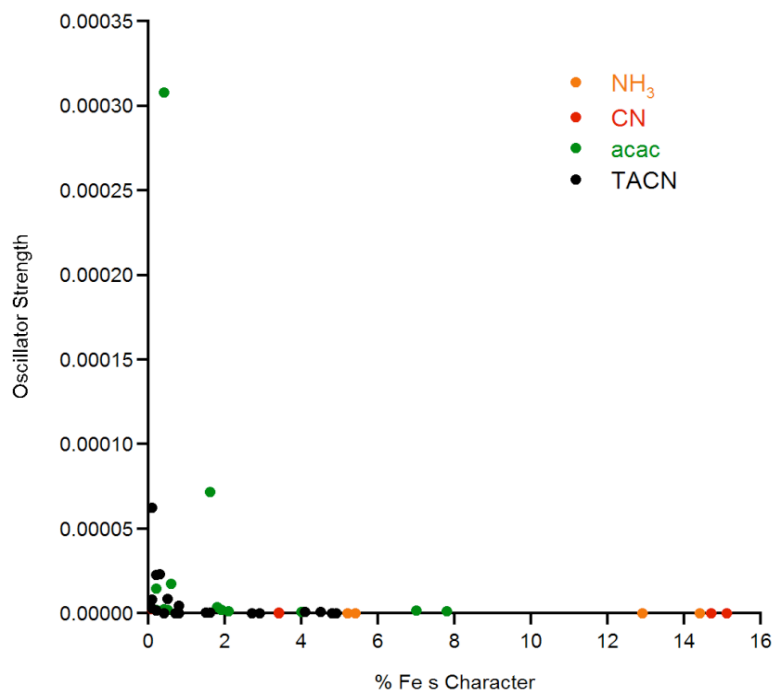
S2. Comparison of N and Fe emission spectra for $\text{Fe}(\text{TACN})_2^{3+}$. Intensity modulation from the perspective of Fe is clearly seen.



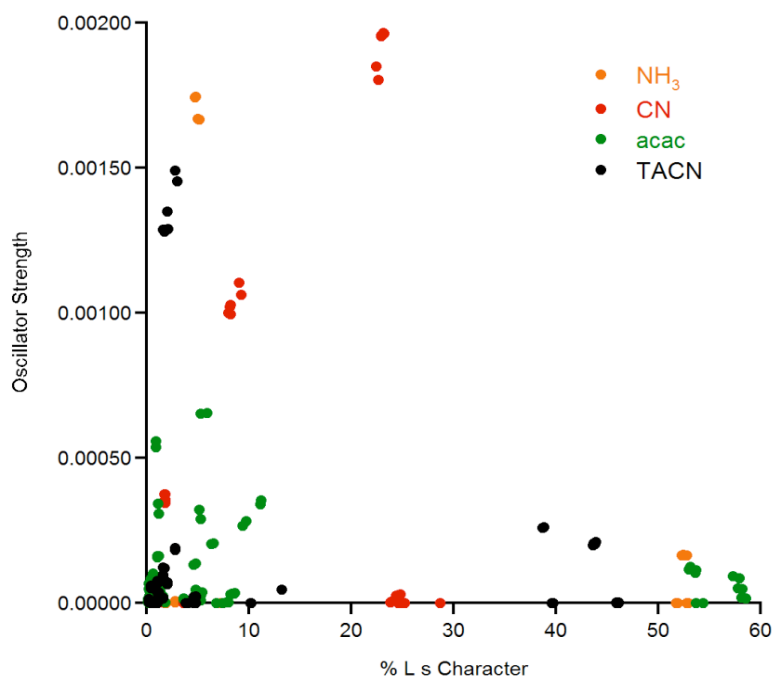
S3. N emission spectra from Q(CN)_6^{n+} showing effect of increase the charge of Q from 0 to +3.



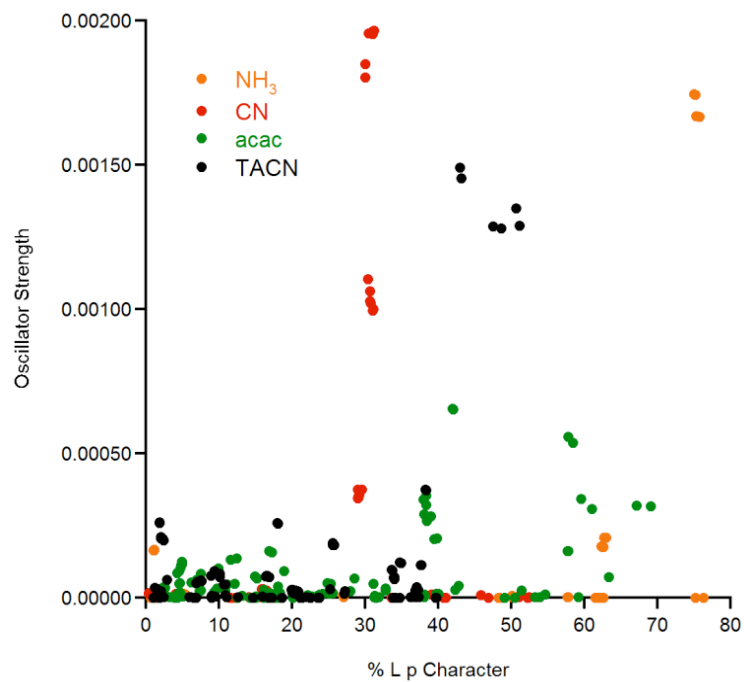
S4. Oxygen emission spectra for $Q(acac)_3^{3-}$, $Q(acac)_3^0$, and $Fe(acac)_3$ that demonstrate the complexity of analyzing this ligand framework. Detailed analysis suggests the same trends apply for these constructs as for the more simple ones analyzed previously.



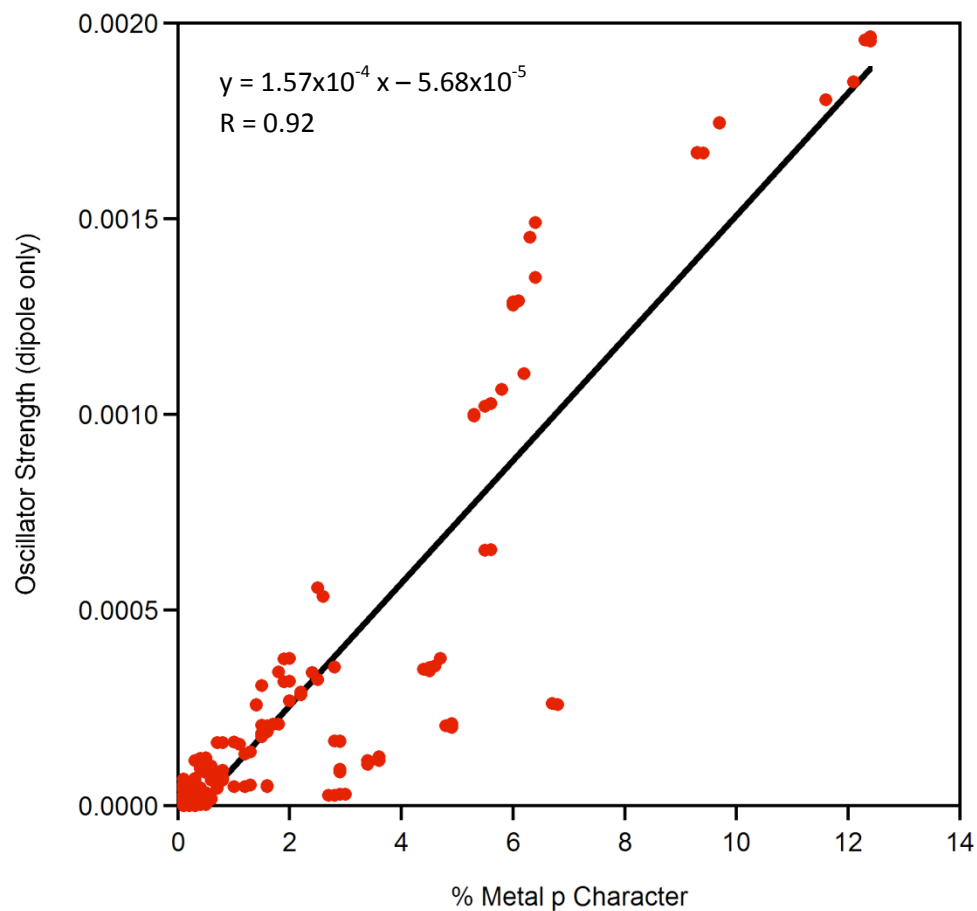
S5. Demonstration that, for the four compounds studied here, there is no correlation between Fe s character and transition intensity (determined from the oscillator strength).



S6. Demonstration that there is no correlation between ligand s character and transition intensity.



S7. Demonstration that there is no correlation between ligand p character and transition intensity.



S8. The correlation between the electric dipole only oscillator strength for the observed transitions and the percent metal p character (shown here) is almost identical to the correlation when the combined electric dipole + magnetic dipole + quadrupole oscillator strength is used.